

The Programme for International Student Assessment has had several runs before the most recent one in 2012. The first PISA assessment was carried out in 2000. The results of each period of assessment take about one year and a half to be analysed. First results were published in November 2001. The release of raw data and the publication of technical report and data handbook only took place in spring 2002. The triennial repeats follow a similar schedule; the process of seeing through a single PISA cycle, start-to-finish, always takes over four years. 470,000 15-year-old students representing 65 nations and territories participated in PISA 2009. An additional 50,000 students representing nine nations were tested in 2010. Every period of assessment focuses on one of the three competence fields of reading, math, science; but the two others are tested as well. After nine years, a full cycle is completed: after 2000, reading was again the main domain in 2009.

Results

PISA 2012

PISA 2009

PISA 2006

PISA 2003

The results for PISA 2003 were released on 14 December 2004. This PISA cycle tested 275,000 15 year-olds on mathematics, science, reading and problem solving and involved schools from 30 OECD member countries and 11 partner countries. Note that for Science and Reading, the means displayed are for "All Students", but for these two subjects (domains), not all of the students answered questions in these domains. In the 2003 OECD Technical Report (pages 208, 209), there are different country means (different than those displayed below) available for students who had exposure to these domains.

PISA 2000

The results for the first cycle of the PISA survey were released on 14 November 2001. 265,000 15 year-olds were tested in 28 OECD countries and 4 partner countries on mathematics, science and reading. An additional 11 countries were tested later in 2002.

Comparison with other studies

The correlation between PISA 2003 and TIMSS 2003 grade 8 country means is 0.84 in mathematics, 0.95 in science. The values go down to 0.66 and 0.79 if the two worst performing developing countries are excluded. Correlations between different scales and studies are around 0.80. The high correlations between different scales and studies indicate common causes of country differences (e.g. educational quality, culture, wealth or genes) or a homogenous underlying factor of cognitive competence. European Economic Area countries perform slightly better in PISA; the Commonwealth of Independent States and Asian countries in TIMSS. Content balance and years of schooling explain most of the variation.

Reception

The results from PISA 2003 and PISA 2006 were featured in the 2010 documentary Waiting for "Superman".

China

Education professor Yong Zhao has noted that PISA 2009 did not receive much attention in the Chinese media, and that the high scores in China are due to excessive workload and testing, adding that it's "no news that the Chinese education system is excellent in preparing outstanding test takers, just like other education systems within the Confucian cultural circle: Singapore, Korea, Japan, and Hong Kong."

Students from Shanghai, China, had the top scores of every category (Mathematics, Reading and Science) in PISA 2009. In discussing these results, PISA spokesman Andreas Schleicher,

Deputy Director for Education and head of the analysis division at the OECD's directorate for education, described Shanghai as a pioneer of educational reform in which "there has been a sea change in pedagogy". Schleicher stated that Shanghai abandoned its "focus on educating a small elite, and instead worked to construct a more inclusive system. They also significantly increased teacher pay and training, reducing the emphasis on rote learning and focusing classroom activities on problem solving."

Schleicher also states that PISA tests administered in rural China have produced some results approaching the OECD average: Citing further, as-yet-unpublished OECD research, Schleicher said, "We have actually done Pisa in 12 of the provinces in China. Even in some of the very poor areas you get performance close to the OECD average." Schleicher says that for a developing country, China's 99.4% enrollment in primary education is "the envy of many countries". He maintains that junior secondary school participation rates in China are now 99%; and in Shanghai, not only has senior secondary school enrollment attained 98%, but admissions into higher education have achieved 80% of the relevant age group. Schleicher believes that this growth reflects quality, not just quantity, which he contends the top PISA ranking of Shanghai's secondary education confirms. Schleicher believes that China has also expanded school access and has moved away from learning by rote. According to Schleicher, Russia performs well in rote-based assessments, but not in PISA, whereas China does well in both rote-based and broader assessments.

Denmark

University of Copenhagen Professor Svend Kreiner, who examined in detail PISA's 2006 reading results, noted that in 2006 only about ten percent of the students who took part in PISA were tested on all 28 reading questions. "This in itself is ridiculous," Kreiner told Stewart. "Most people don't know that half of the students taking part in PISA (2006) do not respond to any reading item at all. Despite that, PISA assigns reading scores to these children."

Finland

The stable, high marks of Finnish students have attracted a lot of attention. According to Hannu Simola the results reflect a paradoxical mix of progressive policies implemented through a rather conservative pedagogic setting, where the high levels of teachers' academic preparation, social status, professionalism and motivation for the job are concomitant with the adherence to traditional roles and methods by both teachers and pupils in Finland's changing, but still quite paternalistic culture. Others advance Finland's low poverty rate as a reason for its success. Finnish education reformer Pasi Sahlberg attributes Finland's high educational achievements to its emphasis on social and educational equality and stress on cooperation and collaboration, as opposed to the competition among teachers and schools that prevails in other nations.

India

Of the 74 countries tested in the PISA 2009 cycle including the "+" nations, the two Indian states came up 72nd and 73rd out of 74 in both reading and mathematics, and 73rd and 74th in science. In Himachal Pradesh, 57.9 percent of 15 year olds in school cannot be distinguished from not having learned any science at all and in TN 43.6 percent all in this category - ten times as many as the USA. The estimate of the fraction of Tamil Nadu or Himachal Pradesh students at level 6 in science proficiency was zero. Their estimate of the fraction at level 5: also zero. Compared to this, about 20 percent of Singapore students reach at least level 5 or 6, while those below level 1 is only 2.8 percent. India's poor performance may not be linguistic as some suggested. 12.87% of US students, for example, indicated that the language of the test differed from the language spoken at home. while 30.77% of Himachal Pradesh students indicated that the language of the test differed from the language spoken at home, a significantly higher percent However, unlike American students, those Indian students with a

different language at home did better on the PISA test than those with the same language. India's poor performance on the PISA test is consistent with India's poor performance in the only other instance when India's government allowed an international organization to test its students and consistent with India's own testing of its elite students in a study titled Student Learning in the Metros 2006. These studies were conducted using TIMSS questions. The poor result in PISA was greeted with dismay in the Indian media. The BBC reported that as of 2008, only 15% of India's students reach high school.

Italy / South Tyrol

In 2003 South Tyrol (Provincia Autonoma di Bolzano / Autonome Provinz Bozen), a predominantly German-speaking province in the north of Italy, took part in the PISA project for the first time in order to have a regional result as an adjudicated region. In the rest of Italy PISA is conducted by INVALSI (Istituto nazionale per la valutazione del sistema educativo di istruzione e di formazione), a formally independent research institution affiliated to the Ministry of Education, whereas in South Tyrol PISA was carried out by the regional Education Authority itself (Intendenza scolastica / Schulamt, since 2018 renamed into Bildungsdirektion), which is part of the South Tyrolean regional government.

At the end of 2004, in the months prior to the announcement of the test results, the regional Education Authority in Bolzano / Bozen downplayed the validity of the PISA assessment and commissioned alternative school evaluations, preparing the public for a mediocre test result. According to the official PISA report 2003, however, South Tyrol seemed to even beat the PISA world champion Finland.

Critique

Right from the beginning, there was scepticism as to how South Tyrol succeeded in outdoing the neighbouring Italian and Austrian provinces. On the front page of its weekend edition for 29/30 January 2005, the South Tyrolean newspaper Neue Südtiroler Tageszeitung published a

harsh critique and revealed that the South Tyrolean Education Authority had secretly eliminated more than 300 students from the 1500 students officially drawn as South Tyrolean test sample by the PISA Consortium, and soon more inconsistencies were to surface:

Lack of independence of the South Tyrolean PISA board: In South Tyrol, PISA was not conducted by a nominally independent body like the local university Freie Universität Bozen or the European Academy EURAC, both of which have ample expertise in the field of education, but by the so-called Pedagogical Institute (Pädagogisches Institut, director: Rudolf Meraner), which was part of the Education Authority, which in its turn was part of the regional government. A few years later the Pedagogical Institute was nominally absorbed into the Education Authority and renamed into Department for Innovation and Counselling (with the same director: Rudolf Meraner).

Exploitation for political ends: In the most influential mass media, the regional PISA results were presented as a triumph of the regional government and the SVP (Südtiroler Volkspartei or South Tyrolean People's Party) ruling party's policy, although the legal framework for all high schools in Italy is a purely national domain. For all the political autonomy granted to South Tyrol, this region still has the same types of schools and follows the same curricula as do all Italian regions. Mass media like the most-read South Tyrolean newspaper, Dolomiten, whose owner Michel Ebner is a prominent party member of the ruling SVP, did not either try to explain why the secondary schools attended by the Italian speaking students in the same Province did considerably worse, although it is the same regional government run by the SVP which is in charge of the Italian and the German school administration.

Harassment of Critics: People who criticized the official PISA results and pointed out violations of the technical rules were officially threatened by the provincial governor, Luis Durnwalder, with libel action for slandering South Tyrol. On 16 March 2006, Durnwalder announced in a press conference that an Austrian teacher would be prosecuted and sued for damages simply

because the teacher, in a letter sent to the Austrian Ministry of Education, had mentioned the fact that the South Tyrolean Education Authority had eliminated 17 percent of the students from the regional sample, thereby rendering invalid the regional PISA result. A year later, however, Mr Durnwalder had to admit that he had never taken legal action against the teacher and that there were no legal proceedings obviously because the critique was correct. It is also noteworthy that the director of the Pedagogical Institute, Rudolf Meraner, and others have constantly deleted the original German Wikipedia article about the South Tyrolean PISA results, replacing it with public government statements.

Regional results deliberately misrepresented as national results: By definition, the PISA result of South Tyrol is a subregional result, which is not fully valid because of the small sample of 1500 students. Such regional results mainly serve documentary purposes and cannot be compared with national results. Nonetheless, the South Tyrolean Education Authority and the regional government repeatedly, and falsely, claimed that the South Tyrolean test results are national, i.e. fully valid results, whereas the neighbouring regions like Tyrol (Austria) and Trentino (Italy), according to South Tyrolean press releases, only had subnational results.

Manipulation of the sample: Elimination of 17 percent of students: For a subnational result, regions had to test 1500 students. Among all regions with a subnational result, worldwide, South Tyrol was the only one that failed to test 1500 students. For reasons never specified, the Education Authority had eliminated 292 students (i.e. 17 percent) from the 1500 students sample, testing only 1208 of the students selected by the PISA consortium. Failing to test all 1500 students, the South Tyrolean Education Authority violated the technical PISA rules and deprived the PISA result even of its limited technical validity as a subnational result. Later on the Education Authority had to admit that it had actually excluded all vocational students, whose performance is generally considered to be inferior to that of high schools, and all students of the third forms of so-called middle schools (*scuole medie* / *Mittelschulen*), who would have been part of the sample only if they repeated a class and were considered

underachievers because normally 15 year old students are enrolled at high schools. From the official PISA report this manipulation can be easily deduced as the difference between total population of 15-year-olds (4.908) and total enrolled population of 15-year-olds at grade 7 or above (4.087). In subsequent PISA assessments, the OECD or rather the PISA consortium did not publish these key figures so that it was no longer possible to ascertain the number of students eliminated from the sample against the rules.

Incorrect figures about Target Population and actual sample: In Italy, according to the figures officially made available by the INVALSI, the number of all 15-year-old students (574.611) paradoxically exceeds the number of all 15-year-old people (561.304). Due to this mistake, it is impossible to establish how many Italian students were actually identified as Italian target population, and it is also impossible to find out which percentage of South Tyrolean students had actually been presented as target population to the PISA consortium. In theory, the Education Authority could have excluded certain types of students from the target population even before the sample of 1500 students was drawn by the PISA consortium, from which the Education Authority later on eliminated 17 percent.

Comparison with similar assessments

The stunning South Tyrolean 2003 PISA results can hardly be reconciled with similar high school evaluations, which were not conducted or influenced by the South Tyrolean Education Authority itself. Three international or national large scale assessment projects painted a gloomy picture of the South Tyrolean students' performance.

Admission Test at Austrian Universities of Medicine (EMS = Eignungstest für das Medizinstudium): Over the last two decades, candidates graduating from South Tyrolean high schools have traditionally scored very badly at the entry examination. The quota system at Austrian Universities of Medicine makes sure that 75 percent of all applicants admitted to Medicine are Austrian nationals or students from South Tyrol, who are considered an Austrian

minority in Italy. Hence, for Austrians and South Tyroleans it is easier to get a place at an Austrian University of Medicine, even with bad scores, than for Germans and other EU nationals because the number of EU students must not exceed 20 percent. Nevertheless, according to a study financed by the Austrian Ministry of Education, not even within the big 75 percent quota, South Tyroleans could get a place because of their mediocre entrance test results. South Tyrolean applicants had an average score of 96,5 whereas e.g. German applicants had an average score of 103,1. Interestingly, South Tyrolean applicants did even worse than Austrian applicants although Italian high school graduates are one year older. What is even worse, South Tyrolean high school graduates show a sharp gender gap with female candidates doing much worse than their male colleagues, which reflects the Italian high school system: Certain types of high schools popular with female students (e.g. *licei linguistici*, *licei scienze umane*) have little mathematics and sciences. The difference between the average female and the average male South Tyrolean score in the 2007 EMS was dramatic: 94,8 versus 100,1 points. Later on the Swiss EMS organisation disallowed the Austrian Universities of Medicine to use its EMS test and the Austrian universities introduced a new entrance test scheme with an extra quota for female students. South Tyrolean mass media did not cover the EMS debacle at all. Within the South Tyrolean parliament, however, there have been debates as to the reasons of the poor performance of South Tyrolean high school students. As a consequence of the EMS debacle, the former South Tyrolean governor, Luis Durnwalder, envisaged a collaboration between the University of Medicine in Innsbruck and the Hospital in Bolzano (Bozen) which should have led to a Euregio Medical School open to all North Tyrolean (i.e. Austrian) and South Tyrolean applicants without any entrance exam. Eventually, the South Tyrolean government adopted a more pragmatic approach, it subsidized, as it were, extra places for South Tyrolean students. This seems to be a breach of the national quota system and the entrance requirements as outlined by Austrian laws. The former South Tyrolean governor, Luis Durnwalder, however, frankly admitted to resort to this

solution when he complained about the high amount of money South Tyrol paid to the University of Medicine in Innsbruck for buying extra places for South Tyrolean students.

German DESI assessment of linguistic skills in German (mother tongue) and English (foreign language): Parallel with the 2003 PISA assessment, the South Tyrolean Education Authority commissioned a second assessment, obviously because the Education Authority expected a mediocre PISA result and had downplayed the importance of PISA. Like PISA, the DESI assessment was conducted against the technical rules because, again, all vocational students attending a vocational school (Berufsschule) and working part-time in a workshop or small firm, i.e. one third of the target population was excluded from the test. The South Tyrolean Education Authority also excluded the whole English test section from DESI, thereby depriving DESI of its main purpose, namely a comparison of German speaking students' language competence in L1 and L2. The South Tyrolean Education Authority argued that in (bilingual) South Tyrol, German speaking students start with Italian as their first foreign language at elementary schools. This explanation, however, lacks credibility because in Germany, as well, many students assessed by DESI had main foreign languages other than English, e.g. French or even Latin (Bavaria), and the amount of weekly classroom teaching in English from the first form at elementary schools is the same in Germany and South Tyrol. In their official report, the German experts responsible for the DESI assessment in South Tyrol generally praised the quality of teaching, but the concrete results in the report reveal drastic shortcomings. For instance, in the semantic field of railway station (Bahnhof), not even one South Tyrolean student with German as his or her mother tongue knew the German word for a signal box (Stellwerk). Instead of marking such errors as errors, however, the German experts resorted to a methodologically disputable assumption. They claimed, without checking, that all words which South Tyrolean students did not know are, by definition, not used in the South Tyrolean variant of German and that these words must be excluded from the test (item bias), thus counting only correct answers. The DESI testers from Germany, however, did not check if the

words excluded from DESI because of item bias actually were unknown. In fact, all words excluded from the South Tyrolean DESI questionnaire are common German words used in South Tyrol as well, i.e. for these words there are no South Tyrolean variants at all. For all the technical exceptions and modifications, the 2003 South Tyrolean DESI result was very disappointing. Only 14 percent of the South Tyrolean high school students came into the best achievement group, whereas in Germany almost half of the 15 year old students belong to this group. On the other hand, one fourth of the South Tyrolean German speaking students came into the lowest achievement group, which in Germany, in spite of all social problems in big towns, comprises only 7 percent of all students.

Italian INVALSI assessments: Traditionally, the annual assessments conducted by the Italian INVALSI in the pre-PISA period painted a sorry picture of the northernmost Italian Regione Autonoma di Trentino e Alto Adige comprising the two autonomous provinces called Trentino (Italian speaking) and South Tyrol (predominantly German speaking). Schools in Trentino and South Tyrol did constantly worse than those in all other North Italian regions and even lagged behind the Italian average, though it is not clear if the sample of German speaking students was representative because, again, the South Tyrolean Education Authority was entitled to eliminate bad schools from the sample, thereby manipulating the validity of the assessment. To some extent, two South Tyrolean anomalies may account for the traditional bad performance of South Tyrolean students in international evaluations. Until recently, no pedagogical qualifications were required for teachers, not even for permanently employed teachers (*insegnanti di ruolo* / *Stammrollenlehrer*) appointed by the Education Authority in Bolzano (Bozen) through a so-called *concorso*. Besides, the admission criteria for the South Tyrolean *concorsi* have always been inconsistent. For example, a South Tyrolean student who studied at an Italian university German as a Foreign Language or Art History, automatically obtained the teaching license for completely unrelated subjects, like History and Latin, at South Tyrolean high schools.

United States

Two studies have compared high achievers in mathematics on the PISA and those on the U.S. National Assessment of Educational Progress (NAEP). Comparisons were made between those scoring at the "advanced" and "proficient" levels in mathematics on the NAEP with the corresponding performance on the PISA. Overall, 30 nations had higher percentages than the U.S. of students at the "advanced" level of mathematics. The only OECD countries with worse results were Portugal, Greece, Turkey, and Mexico. Six percent of U.S. students were "advanced" in mathematics compared to 28 percent in Taiwan. The highest ranked state in the U.S. (Massachusetts) was just 15th in the world if it was compared with the nations participating in the PISA. 31 nations had higher percentages of "proficient" students than the U.S. Massachusetts was again the best U.S. state, but it ranked just ninth in the world if compared with the nations participating in the PISA.

Comparisons with results for the Trends in International Mathematics and Science Study (TIMSS) appear to give different results—suggesting that the U.S. states actually do better in world rankings. This can likely be traced to the different material being covered and the United States teaching mathematics in a style less harmonious with the "Realistic Mathematics Education" which forms the basis of the exam. Countries that commonly use this teaching method score higher on PISA, and less highly on TIMSS and other assessments.

Poverty

Stephen Krassen, professor emeritus at the University of Southern California, and Mel Riddile of the NASSP attributed the relatively low performance of students in the United States to the country's high rate of child poverty, which exceeds that of other OECD countries. However, individual US schools with poverty rates comparable to Finland's (below 10%), as measured by reduced-price school lunch participation, outperform Finland; and US schools in the 10–24%

reduced-price lunch range are not far behind.

Reduced school lunch participation is the only available intra-poverty indicator for US schoolchildren. In the United States, schools in locations in which less than 10% of the students qualified for free or reduced-price lunch averaged PISA scores of 551 (higher than any other OECD country). This can be compared with the other OECD countries (which have tabled figures on children living in relative poverty):

Sampling errors

In 2013 Martin Carnoy of the Stanford University Graduate School of Education and Richard Rothstein of the Economic Policy Institute released a report, "What do international tests really show about U.S. student performance?", analyzing the 2009 PISA data base. Their report found that U.S. PISA test scores had been lowered by a sampling error that over-represented adolescents from the most disadvantaged American schools in the test-taking sample. The authors cautioned that international test scores are often "interpreted to show that American students perform poorly when compared to students internationally" and that school reformers then conclude that "U.S. public education is failing." Such inferences, made before the data has been carefully analyzed, they say, "are too glib" and "may lead policymakers to pursue inappropriate and even harmful reforms."

Carnoy and Rothstein observe that in all countries, students from disadvantaged backgrounds perform worse than those from advantaged backgrounds, and the US has a greater percentage of students from disadvantaged backgrounds. The sampling error on the PISA results lowered U.S. scores for 15-year-olds even further, they say. The authors add, however, that in countries such as Finland, the scores of disadvantaged students tends to be stagnant, whereas in the U.S the scores of disadvantaged students have been steadily rising over time, albeit still lagging behind their those of their more advantaged peers. When the figures are adjusted for social class, the PISA scores of all US students would still remain behind those of

the highest scoring countries, nevertheless, the scores of US students of all social backgrounds have shown a trajectory of improvement over time, notably in mathematics, a circumstance PISA's report fails to take into account.

Carnoy and Rothstein write that PISA spokesman Schleicher has been quoted saying that "international education benchmarks make disappointing reading for the U.S." and that "in the U.S. in particular, poverty was destiny. Low-income American students did (and still do) much worse than high-income ones on PISA. But poor kids in Finland and Canada do far better relative to their more privileged peers, despite their disadvantages" (Ripley 2011). Carnoy and Rothstein state that their report's analysis shows Schleicher and Ripley's claims to be untrue. They further fault the way PISA's results have persistently been released to the press before experts have time to evaluate them; and they charge the OECD reports with inconsistency in explaining such factors as the role of parental education. Carnoy and Rothstein also note with alarm that the US secretary of education Arne Duncan regularly consults with PISA's Andreas Schleicher in formulating educational policy before other experts have been given a chance to analyze the results. Carnoy and Rothstein's report (written before the release of the 2011 database) concludes: We are most certain of this: To make judgments only on the basis of national average scores, on only one test, at only one point in time, without comparing trends on different tests that purport to measure the same thing, and without disaggregation by social class groups, is the worst possible choice. But, unfortunately, this is how most policymakers and analysts approach the field.

The most recent test for which an international database is presently available is PISA, administered in 2009. A database for TIMSS 2011 is scheduled for release in mid-January 2013. In December 2013, PISA will announce results and make data available from its 2012 test administration. Scholars will then be able to dig into TIMSS 2011 and PISA 2012 databases so they can place the publicly promoted average national results in proper context. The analyses

we have presented in this report should caution policymakers to await understanding of this context before drawing conclusions about lessons from TIMSS or PISA assessments.

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